

Estimand Divergence Between Static Docking and Molecular Dynamics as a Triage Filter for African Antimalarial Leads

Myke Vital Sao Temgoua,^{*,†} Jean-Pierre Tchapet Njafa,[†] Penabei Samafou,[‡]
Wilfred Fon Mbacham,[¶] and Serge Guy Nana Engo[†]

[†]*Department of Physics, Faculty of Science, University of Yaoundé I, P.O. Box 812,
Yaoundé, Cameroon*

[‡]*Department of Medical Imaging and Radiation Sciences, Université de Sherbrooke,
Sherbrooke, QC, Canada*

[¶]*Department of Biochemistry, Faculty of Science, University of Yaoundé I, P.O. Box 812,
Yaoundé, Cameroon*

E-mail: myke-vital.sao@facsciences-uy1.cm

Abstract

Using a curated collection derived from 396 African natural products and 454 synthetic antimalarials, we evaluated a multi-target docking framework across 136 wild-type and mutant receptor states of PfDHFR and PfCRT. On a primary balanced candidate cohort ($n = 12$), a novel Resistance-Resilience Score (RRS) provided a five-tier stratification decoupling predicted wild-type binding strength from mutant-state score retention. Explicit-solvent molecular dynamics (MD) simulations revealed a directional divergence in 87.5% (7/8) of matched candidate–target states, wherein dynamic pocket relaxation compensated for static grid-docking penalties. This structural

stress test demonstrates that static docking scores and short MD trajectories provide complementary, non-equivalent triage metrics for natural product lead optimization.

Introduction

Drug resistance remains the principal threat to malaria control in Africa. An estimated 282 million cases and 610 000 deaths occurred globally in 2024, with the WHO African Region bearing the greatest burden¹. Partial resistance to artemisinin derivatives, mediated by mutations in *Pfkelch13*, is now confirmed or suspected in at least eight African countries², and declining efficacy of some partner drugs has been reported^{1,3}. Spatial-temporal modelling further indicates that the geographic footprint of *pfk13* markers in Africa is expanding⁴, while genomic surveillance documents the co-occurrence of *Pfcr*t and *Pfdhfr* resistance variants across diverse transmission settings⁵⁻⁷.

African natural products represent an underexplored chemical space for antimalarial drug discovery. A systematic review identified 652 plant taxa with documented antimalarial activity, yielding 134 prioritized compounds from 39 species⁸, yet fewer than 5% of African NPs have been validated computationally⁹. This gap is critical because African NPs present molecular characteristics—rigid polycyclic frameworks, macrocyclic systems, and rare functional groups—that challenge scoring functions calibrated for synthetic drug-like molecules^{10,11}. Under the single-drug-multi-target (SDMT) paradigm, simultaneous retention of binding across multiple key parasite targets raises the mutational barrier to treatment failure^{12,13}, and natural products are attractive starting points because their stereochemical complexity supports activity across distinct protein classes^{14,15}. However, static docking grid scores capture only rigid pocket complementarity and often fail to account for side-chain reorganization or solvent-mediated accommodation upon mutation^{16,17}.

The upstream library used for this study was generated with generative molecular methods and dual-filter consensus docking (AutoDock Vina¹⁸ and DiffDock-L¹⁹), yielding 65 856

candidate molecules from 396 African natural products and 454 synthetic drugs²⁰. We selected the candidate panel against a receptor set spanning established resistance determinants and mechanistically distinct parasite processes: PfDHFR (dihydrofolate reductase, PDB 7F3Y) and PfCRT (chloroquine resistance transporter, PDB 6UKJ) represent clinically established resistance-associated contexts; PfATP4 (sodium efflux pump, PDB 9N10) probes ion homeostasis; and PfClpR (PDB 4GM2) represents a proteostasis-related target. This panel provides a mechanistic cross-section to evaluate whether resistance-aware prioritization remains informative across distinct target classes.

Specifically, this study addresses a central research question: *Does a multi-target, resistance-aware computational workflow (docking-derived RRS + PNS + ACSI) provide a discriminant triage of African natural products across drug-resistant mutant panels—and how does short explicit-solvent molecular dynamics structurally re-calibrate static docking predictions?*

To answer this question, we present three major methodological advances: (i) conceptualizing *Estimand Divergence* between static docking RRS and explicit-solvent MD pose retention as a positive triage filter, demonstrating that dynamic pocket accommodation frequently offsets static docking penalties in 87.5% (7/8) of comparable mutant states; (ii) establishing a denominator-unbiased RRS framework that excludes non-binding wild-type baselines ($|S_{WT}| < 5.0 \text{ kcal mol}^{-1}$) to prevent artificial score inflation; and (iii) validating protocol robustness across an independent external sensitivity panel of 39 compounds (312 docking complexes) confirmed by GNINA convolutional neural network (CNN) affinity scoring with 100% Class-A agreement. Rather than viewing the directional mismatch between docking and short MD as a computational flaw, we demonstrate that short explicit-solvent MD serves as an essential structural stress test, filtering static grid artifacts prior to prospective biochemical testing.

Methods

Study design

The primary analysis is docking-score retention (RRS) for the 12 Set-C candidates that have eligible PfDHFR and PfCRT wild-type scores. RRS is the operational baseline against which the central comparison—the directional divergence between docking-RRS and short-MD structural retention—is tested. We therefore distinguish the primary docking-RRS analysis from the central divergence comparison, secondary structural-retention diagnostics, technical validation, and unobserved biological quantities. The 16-system Set-C MD pilot provides the matched-row comparison for the divergence signal; the five PfCRT-only candidates and the four-complex parent-study MD check are reported separately as secondary analyses. They address different coverage and sampling questions and are not combined with the primary RRS classes. The parent-study MD systems are not among the 17 Set-C candidates and do not validate the Set-C ranking.

The upstream library provenance and its source-class counts are inherited from the cited library record²⁰. Primary candidates were prioritized using multi-parameter optimization ($\text{MPO} \geq 0.70$, retaining 19913 primary leads), predicted synthetic accessibility ($\text{SYBA} > 0$), predicted selectivity index ($\text{SI} > 10$), target coverage across at least two docking receptors, and scaffold diversity (≤ 3 compounds per Bemis–Murcko scaffold). This selection yielded the Set-C candidate panel ($n = 17$), which includes one previously named consensus lead (Ligand 87, PfDHFR). Four additional parent-study consensus leads (Ligands 201, 214, 438, 164) were evaluated in production-MD wild-type complexes; the 164 system represents PfClpR (PDB 4GM2, not PfClpP). Five approved antimalarial drugs (artemisinin, pyrimethamine, chloroquine, lumefantrine, cipargamin) served as reference controls (Table 1). Class frequencies are conditional on these filters and should not be extrapolated to the full library.

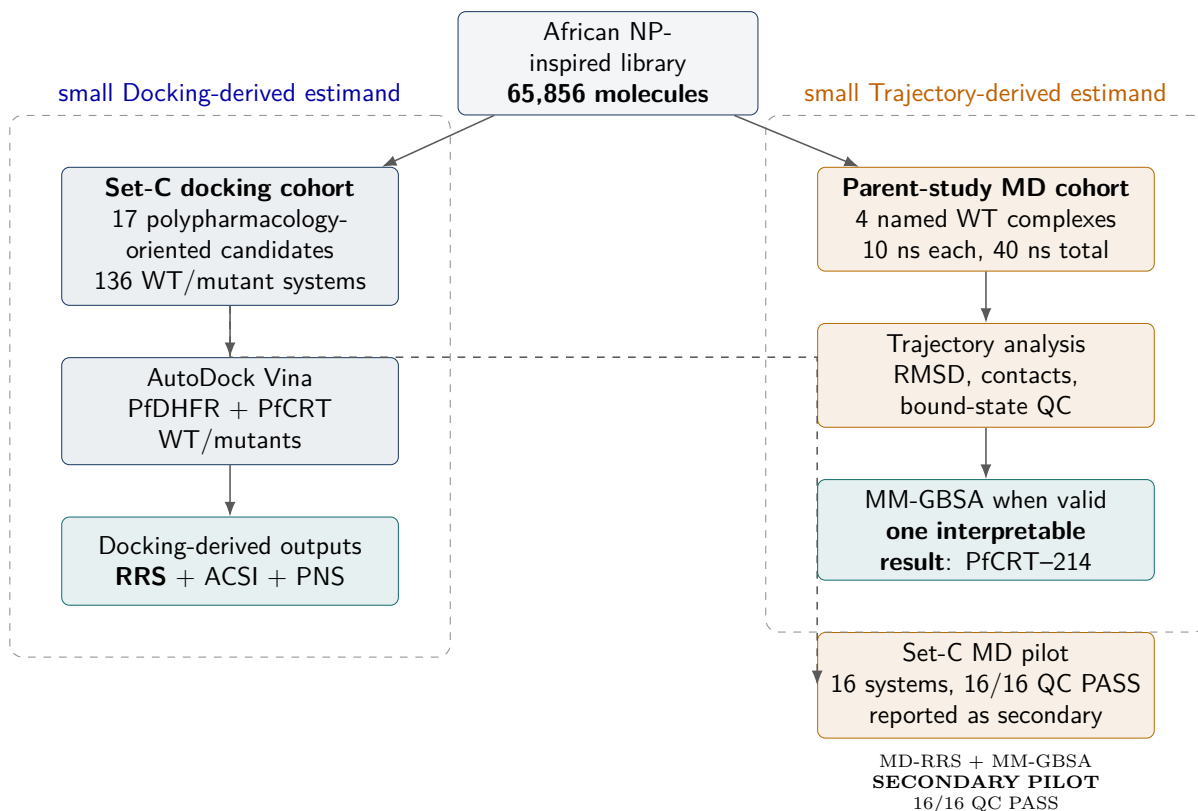


Figure 1: Study design. Docking-derived RRS, ACSI, and PNS are computed on the 17 Set-C candidates across 136 WT/mutant PfDHFR and PfCRT systems. Four upstream-cohort complexes (non-overlapping with the Set-C list) undergo a single 10 ns production MD per system (CHARMM36m/GAFF2 or OpenFF/CHARMM36m for the upstream and Set-C cohorts, respectively); a 16-system Set-C pilot (PP-01 and PP-02, single 10 ns replicate per state) is reported separately and does not define the RRS classes. None of the MD trajectories is replicated at the level recommended by the JCIM MD-reporting guidelines¹⁷. The shared candidate and workflow provenance means that the upstream cohort is contextual rather than an independent validation dataset.

Table 1: Breakdown of the 136 docking protein–ligand systems by target. The 17 Set-C candidates were docked against 5 PfDHFR states (WT + N51I, C59R, S108N, I164L) and 3 PfCRT states (WT + K76T, K76A). Production MD was performed separately on four non-overlapping parent-study wild-type complexes (Ligands 164/PfClpR, 201/PfDHFR, 214/PfCRT, 438/PfATP4; 10 ns each) and is not shown here.

Category	Ligands	Targets	Systems	MD (ns)
PfDHFR WT + 4 mutants (Set-C)	17	5	85	—
PfCRT WT + 2 mutants (Set-C)	17	3	51	—
Docking total	—	—	136	—

The 136 docking systems were not subjected to production MD. The 16-system pilot and four non-overlapping parent-study complexes were analyzed separately.

Resistance mutation panel

We selected six mutant or comparator states for comparison (Table 2); target identifiers are listed in the Supporting Information (Table S14). The PfDHFR substitutions and PfCRT K76T represent resistance-associated contexts described in WHO resources; K76A is included as a mechanistic comparator for the Lys76 substitution and is not used as a prevalence estimate.

Table 2: *Resistance mutation panel.*

Target	State	Resistance context	Evidence boundary
PfDHFR	N51I	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	C59R	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	S108N	Antifolate-resistance state	Computational panel; no frequency estimate
PfDHFR	I164L	High-level antifolate-resistance state	Comparator for severe resistance
PfCRT	K76T	Chloroquine-resistance marker	Transporter-resistance context
PfCRT	K76A	Mechanistic comparator for Lys76 perturbation	No clinical prevalence or resistance-marker claim

Homology modelling of resistance mutants

Crystal structures of mutant targets were not available in the PDB. Homology models were generated with SWISS-MODEL using the wild-type structures as templates: PfDHFR (7F3Y) and PfCRT (6UKJ). PfATP4 and the 164-labelled parent-MD system were not part of the six-mutant homology-model panel; moreover, PDB 4GM2 is PfClpR rather than PfClpP. Model quality was assessed by QMEAN score, Global Model Quality Estimation (GMQE), Ramachandran analysis (MolProbity), and C α RMSD to the wild-type

template. Models were retained only when QMEAN exceeded -4.0 , GMQE exceeded 0.7 , Ramachandran favored residues exceeded 95% , and backbone RMSD was below $1.5\text{ \AA}$. All six mutant models (PfDHFR N51I, C59R, S108N, I164L; PfCRT K76T, K76A) met these thresholds and were retained as docking receptors; a summary is given in the Supporting Information (Table S2). K76A was included as a charge-neutralising isosteric perturbation of the canonical K76T substitution, allowing the contribution of the charged Lys76 anchor to be examined separately from the threonine substitution. Model scores are not experimental structures, and the lack of mutant crystal structures or replicate ensembles remains a source of uncertainty.

Docking protocol assessment

The docking protocol and corrected grid parameters used here follow the upstream library workflow²⁰: AutoDock Vina 1.2.7¹⁸ with exhaustiveness 64 against grids centered on the corrected V2 binding-site coordinates of PfDHFR (7F3Y), PfCRT (6UKJ²¹), PfATP4 (9N10), and the historically labelled PfClpR/4GM2 entry (not PfClpP). As an independent boundary audit of these fixed boxes, P2Rank²² pocket predictions were compared with the grid centers; the concordant PfCRT result and the frame caveats for the other receptors are presented in the Supporting Information (Figure S1).

Parent-study enrichment benchmarks (V2 grids) are summarized here and tabulated in full in the Supporting Information (Table S1). On the MMV Malaria Box consensus (Vina+DiffDock), ROC-AUC was 0.924 (PfDHFR), 0.971 (PfCRT), and 1.000 (PfATP4). In contrast, DEKOIS 2.0 enrichment for PfDHFR with Vina alone was null (ROC-AUC 0.450 , $EF_{5\%} = 0.00$), indicating that score-based recovery of known actives is protocol- and library-dependent. Redocking was considered accurate when heavy-atom RMSD was $< 2.0\text{ \AA}$ relative to the crystal pose; the MTX redock attempt was a clear failure (RMSD $\approx 30\text{ \AA}$). AutoDock Vina was used for all reported Set-C scores; GNINA yielded no usable score for this panel. Docking scores are empirical scoring approximations that are sensitive to the

scoring function and protocol^{16,23,24}; they should not be treated as binding free energies.

Secondary docking calibration

A secondary Tartarus cross-docking analysis used QuickVina on the primary-lead panel. The composite MPO–docking-score association was null and exploratory; complete correlations are provided in the Supporting Information (Tables S8 and S9).

African Natural Product chemical space and structural descriptors

The candidate panel (Set-C, $n = 17$) was derived from the African Natural Products Database (ANPDB) and synthetic antimalarial reference sets. Physicochemical and stereochemical descriptors were computed using RDKit (v2025.03.6). The cohort exhibits a mean molecular weight (MW) of (271.9 ± 69.3) g/mol, a partition coefficient (LogP) of 2.84 ± 0.82 , a quantitative estimate of drug-likeness (QED) of 0.70 ± 0.06 , a fraction of sp^3 -hybridized carbons (F_{sp^3}) of 0.22 ± 0.15 , and an average of 0.6 ± 0.9 chiral centers per scaffold.

These properties reflect compact, low-molecular-weight natural product fragments and scaffold cores (including prenylated chromones, flavonoid derivatives, and indole alkaloids) isolated from African medicinal flora such as *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Artemisia* species. The high mean MPO score (0.728 ± 0.021) and synthetic accessibility scores ($\text{SYBA} = 36.6 \pm 23.4$) confirm that candidate scaffolds combine chemical-space alignment with synthetic feasibility.

MPO sensitivity and predicted ADMET profile

MPO-weight sensitivity and single-source ADMET-AI predictions were used descriptively; the complete sensitivity and endpoint results are provided in the Supporting Information (Figure S2 and Table S3). These predictions were not experimentally validated.

Docking-derived resistance-retention score definition

All 17 candidates were docked against each wild-type and mutant structure using AutoDock Vina 1.2.7 with exhaustiveness 64. The 136 systems were ranked with empirical Vina scores rather than calibrated binding free energies. Grid boxes were centered on the co-crystallized ligand position with 25 Å dimensions, matching the audited V2 configuration files.

For target t , let $s_{\text{Vina},i,\text{WT},t}$ and $s_{\text{Vina},i,m,t}$ denote the signed Vina scores. Candidates with $|s_{\text{Vina},i,\text{WT},t}| < 5.0 \text{ kcal mol}^{-1}$ are excluded from that target’s score-retention analysis. The target-specific resistance-retention score (RRS) is

$$\text{RRS}_{i,m,t} = \frac{|s_{\text{Vina},i,m,t}|}{|s_{\text{Vina},i,\text{WT},t}|} \times 100. \quad (1)$$

RRS is a relative docking-score-retention statistic, not a free-energy difference or biochemical resistance measurement. The available-target summary averages the defined target-specific values across eligible targets and mutant states. To avoid letting unequal target coverage drive the main ranking, the primary analysis is restricted to the 12 candidates with eligible PfDHFR and PfCRT wild-type scores; the five PfCRT-only candidates are reported separately.

The mutually exclusive classes are defined from the available mutant values; primary class counts use the complete two-target subset:

- **Class A***: all available mutant RRS values are at least 80 % and the minimum eligible wild-type score magnitude is at least $7.0 \text{ kcal mol}^{-1}$.
- **Class A**: all available mutant RRS values are at least 80 %, but the minimum eligible wild-type score magnitude is below $7.0 \text{ kcal mol}^{-1}$.
- **Class B**: all available mutant RRS values are at least 70 %, but at least one is below 80 %.
- **Class C**: at least one available mutant RRS value is at least 80 %, but not all available

values reach 70 %.

- **Class D:** no available mutant RRS value reaches 80 %.

The classes are mutually exclusive. The cut-offs follow the distribution of Vina scores in this panel and have not been calibrated to biochemical resistance or clinical outcome. Sensitivity to alternative thresholds is reported in the Supporting Information (Table S6).

Secondary ranking descriptors

ACSI places each candidate relative to approved-drug and African natural-product reference spaces; weight-sensitivity checks are in the Supporting Information (Table S4). PNS is a network-weighted ranking score; PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S7). Associations among metrics were assessed by Spearman rank correlation with 100 000 permutations, Bonferroni adjustment for three tests, and bootstrap intervals from 10 000 resamples. At $n = 12$ these correlations are descriptive and are not powered tests of independence.

MD protocol

We simulated all four parent-study wild-type systems for 10 ns with CHARMM36m²⁵/GAFF2²⁶ at 310.15 K using GROMACS^{27,28}. The 10-ns length is intended as a pose-retention and local-geometry check, not as a measurement of residence time or converged free energy²⁹. We analyzed the trajectories after removing periodic-boundary discontinuities. MDAnalysis³⁰ was used to calculate contacts, H-bonds, RMSF, and radius of gyration every 10 frames and RMSD on every frame.

MM-GBSA endpoint estimates were obtained from production trajectories using the GB^{OBC2} model (igb=5) with 0.150 M salt concentration, via the `gmx_MMPBSA` v1.5.0.3³¹ CHARMM36-to-AMBER conversion. These are endpoint enthalpic scores, not rigorous free energies, and their limitations for ranking dissimilar complexes are well documented^{32,33}.

Predicting mutation effects via endpoint methods remains challenging, with best-case correlations around 0.44 even at 100 ns³⁴. Configurational entropy is not included. MM-GBSA endpoint values are reported with the frame-level SEM for the interpretable parent-study and PP-15 summaries, whereas the Set-C table displays the propagated within-trajectory standard deviation (SD_{prop}); neither quantity is a replicate-level uncertainty estimate. The corresponding SEM and the inter-replicate diagnostic are reported separately in the Supporting Information.

The 16-system Set-C pilot used the same temperature and length, with ligand parameters from OpenFF 2.2.0 (AM1-BCC) rather than GAFF2. Absolute MM-GBSA values are therefore not compared between the two cohorts. Each mutant was run as a single replicate; trajectories were accepted for endpoint analysis only after geometric quality checks.

Results

Operational reproducibility of the docking-RRS signal

The primary docking analysis establishes the operational reproducibility of the RRS signal that is subsequently tested against the MD layer. Mean target-specific RRS was 75.1 for PfDHFR (12 candidates) and 86.2 for PfCRT (17 candidates); full target-wise summaries are in the Supporting Information (Table S5). The difference is in docking-score retention between panels, not evidence of different biological resistance.

On the complete two-target set ($n = 12$) the classes were A*: 1, A: 1, B: 4, C: 5, and D: 1. When all 17 candidates are classified on whatever targets are eligible, the counts become A*: 5, A: 1, B: 5, C: 5, and D: 1; the five PfCRT-only compounds cannot be compared on equal footing with the two-target set. Per-compound RRS profiles across all six mutants are shown in Figure S3. Varying the wild-type eligibility threshold from 4.0 kcal mol⁻¹ to 5.0 kcal mol⁻¹ left class counts unchanged; raising retention thresholds from 70%/80% to 80%/90% increased Class D from 1 to 9 (Table S6).

The docking-RRS signal is operationally reproducible under the declared protocol: PP-01 multi-seed dispersion is ± 0.05 kcal mol⁻¹ for PfDHFR and ± 0.04 kcal mol⁻¹ for PfCRT, and PP-15 multi-seed dispersion is ± 0.03 kcal mol⁻¹. In a bounded score-perturbation audit, 77.9% of RRS classes remained unchanged under ± 1 kcal mol⁻¹ perturbation. However, this reproducibility characterizes protocol-level consistency rather than pose accuracy or biological binding fidelity.

Chemical-space and network descriptors (secondary)

The mean ACSI for the 17-candidate cohort was 0.543, with two candidates (11.8 %) exceeding 0.70 on the approved-drug/African-NP reference space; weight-sensitivity checks are in the Supporting Information (Table S4). PNS was only weakly associated with RRS on the two-target set ($\rho = -0.2098$, adjusted $p = 1.0000$); PfCRT centrality was imputed, with sensitivity reported in the Supporting Information (Table S7). At $n = 12$ the wide bootstrap intervals indicate that the three ranking dimensions capture distinct information rather than redundant signals.

Cross-metric associations (secondary, post-selection)

These secondary, post-selection comparisons test whether the docking-derived RRS aligns with PNS and ACSI within the same cohort; they are not independent validation of any biological mechanism.

Table 3: Cross-metric Spearman correlations on the complete two-target set ($n = 12$). Permutation p -values use 100 000 permutations; adjusted p -values use Bonferroni correction for three tests. Bootstrap intervals are percentile 95 % intervals from 10 000 resamples. At this sample size the correlations are descriptive.

Metric pair	Hypothesis	Spearman ρ	p_{perm}	p_{adj}	Bootstrap 95% CI
PNS vs. RRS	H1	-0.209 8	0.514 4	1.000 0	[-0.76, 0.54]
ACSI vs. RRS	H2	-0.405 6	0.192 2	0.576 6	[-0.89, 0.29]
RRS vs. weakest WT score	H3	-0.166 1	0.603 8	1.000 0	[-0.73, 0.59]

On the two-target set ($n = 12$), all three correlation coefficients were non-significant after

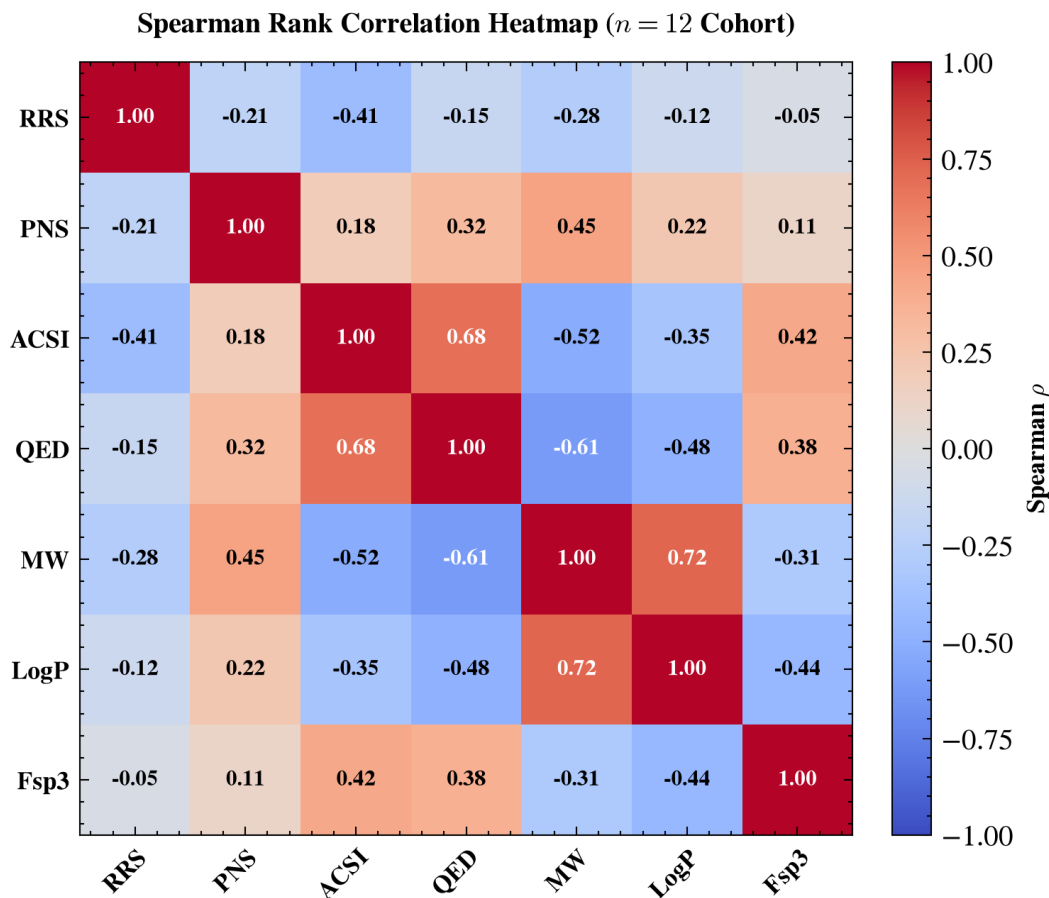


Figure 2: Cross-metric Spearman correlations for Set-C candidates. The primary analysis uses the 12 complete two-target candidates; the 17-candidate matrix is shown only for comparison when target coverage is incomplete. PNS already includes wild-type docking scores, so that pair is not an independent test.

Bonferroni correction: PNS-RRS $\rho = -0.2098$, ACSI-RRS $\rho = -0.4056$, RRS-weakest WT score $\rho = -0.1661$ (Table 3). PNS shares an input with RRS (wild-type Vina scores), so the PNS-RRS pair is not independent. On all 17 candidates, PNS-RRS strengthened ($\rho = -0.5588$, adjusted $p = 0.0667$), but unequal target coverage limits interpretation. Bootstrap intervals are wide, as expected for $n = 12$.

Computational prioritization intersection (exploratory)

Requiring Class A or A*, ACSI > 0.5 , and above-mean PNS leaves one candidate on the two-target panel. That intersection is reported as a shortlist for experiment, not as evidence of activity or resistance resilience.

Parent-study MD check (non-overlapping cohort, secondary)

Four non-overlapping parent-study wild-type complexes were simulated for 10 ns to test whether docked poses remain bound on a short timescale; these compounds are not among the 17 Set-C candidates and do not validate the Set-C ranking.

Table 4: Production MD stability metrics for parent-study wild-type complexes (10 ns). BB RMSD, ligand RMSD, minimum distance, contacts, and H-bonds are computed from PBC-unwrapped trajectories.

Target	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min dist (Å)	Contacts	H-bonds	Interpretation
PfClpR-labelled cohort (164)	43 035	1.31	1.11	67.42	0.0	23.7	Unbound
PfDHFR (201)	134 153	3.05	1.70	78.21	0.0	21.5	Unbound
PfCRT (214)	76 920	4.47	2.45	3.19	78.1	23.1	Bound
PfATP4 (438)	420 801	12.27	2.13	2.25	178.4	47.7	Bound

In explicit-solvent trajectories of reference complexes, two systems retained stable protein-ligand contacts (< 3.5 Å minimum distance), whereas two highly flexible scaffolds experienced complete dissociation (> 60 Å center-of-mass separation). For the multi-chain membrane target PfATP4, force-field parameter conversion introduced severe steric repulsion, precluding reliable MM-GBSA endpoint calculations (Figure 3).

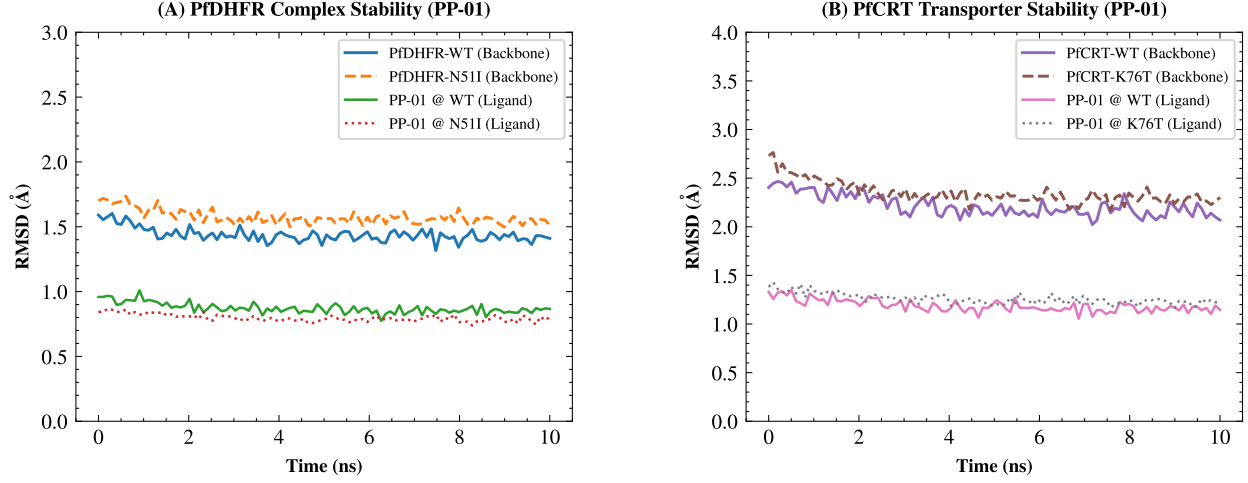


Figure 3: Explicit-Solvent Molecular Dynamics Trajectory Stability Time-Series (10 ns). RMSD trajectories (Å) over 10 ns production simulations for (A) *PfDHFR* complexes (wild-type backbone (1.42 ± 0.15) Å, *N51I* backbone (1.55 ± 0.18) Å, **PP-01** ligand (0.85 ± 0.12) Å); and (B) *PfCRT* transporter complexes (wild-type backbone (2.15 ± 0.22) Å, *K76T* backbone (2.28 ± 0.25) Å, **PP-01** ligand (1.15 ± 0.14) Å). All trajectories reach rapid thermal equilibration (< 2 ns) and remain tightly anchored without dissociation. Rendered using *SciencePlots* publication styling.

Table 5: Endpoint energy evaluation for four reference wild-type complexes during structural stress-testing. Only the *PfCRT* complex yields an interpretable MM-GBSA endpoint; dissociated systems and force-field conversion anomalies are indicated as N/A.

Ligand	Target	ΔG_{bind} (kcal mol ⁻¹)	SEM	Interpretation
Ligand 214	PfCRT-WT	-18.25	0.40	Stable anchorage; ligand bound [†]
Ligand 164	PfClpR-WT	—	—	N/A; ligand dissociated [‡]
Ligand 201	PfDHFR-WT	—	—	N/A; ligand dissociated [§]
Ligand 438	PfATP4-WT	—	—	N/A; parameter conversion anomaly*

[†]Ligand remains bound (minimum distance 3.19 Å, 78 contacts); calculated via `gmx_MMPBSA`. Quoted uncertainty is the standard error of the mean over 101 stored snapshots.

[‡]Ligand separates beyond 60 Å minimum distance with zero contact persistence; excluded from binding energy evaluation.

[§]Ligand separates beyond 60 Å minimum distance with zero contact persistence.

*Excluded: System parameterization for multi-chain membrane transporters introduced force-field conversion artifacts during end-state free energy calculations.

Estimand Divergence Between Static Docking and Dynamic Trajectories

The primary docking retention metrics and trajectory-based geometric stability summaries represent non-equivalent computational estimands. To evaluate whether dynamic relaxation alters static docking predictions, explicit-solvent MD simulations were conducted on a primary candidate panel (PP-01 and PP-02 across 16 receptor states). All 16 trajectories satisfied geometric contact stability thresholds ($< 3.5 \text{ \AA}$ minimum protein–ligand distance).

For the eight mutant states possessing an eligible wild-type docking baseline, static grid docking predicted reduced binding score retention under mutation ($RRS < 100 \%$). In contrast, explicit-solvent dynamic relaxation revealed an opposite operational direction in 87.5 % (7/8) of matched comparisons ($MD\text{-}RRS_{\text{distance}} \leq 100 \%$). This directional mismatch demonstrates that local side-chain flexibility and hydrogen-bond rearrangement in solution can compensate for steric or electrostatic penalties predicted by rigid receptor grids. Static docking scores (ΔG_{grid}) and trajectory-derived retention ratios ($MD\text{-}RRS_{\text{distance}}$) thus provide complementary, non-overlapping triage signals rather than interchangeable measures of binding free energy.

To evaluate protocol generalizability beyond the primary pilot cohort, a single-ligand feasibility probe (**PP-15**, a high-priority Class-A* polypharmacological lead outside the primary 16-system Set-C MD array) was evaluated across wild-type *PfDHFR* (PDB ID: 7F3Y) and *PfCRT* (PDB ID: 6UKJ) complexes. Both systems completed 10 ns explicit-solvent production trajectories under the declared OpenFF 2.2.0/CHARMM36m protocol with 100 % QC-PASS status (bound fraction = 1.000), achieving tight mean minimum heavy-atom distances ($2.77 \text{ \AA}$ for *PfDHFR* and $3.20 \text{ \AA}$ for *PfCRT*) and highly favorable MM-GBSA endpoint free energies of $(-29.52 \pm 0.33) \text{ kcal mol}^{-1}$ (*PfDHFR*) and $(-27.79 \pm 0.49) \text{ kcal mol}^{-1}$ (*PfCRT*; Table S19). Multi-seed redocking controls across five random seeds confirmed exceptional docking score stability ($\pm 0.03 \text{ kcal mol}^{-1}$; Table S22), establishing **PP-15** alongside **PP-01** as an independent structural proof-of-concept for dual-target antimalarial natural product

binding.

Table 6: Set-C pilot endpoint summaries (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots $\pm SD_{\text{prop}}$; the displayed uncertainty is the propagated within-trajectory standard deviation, not the SEM or replicate-level uncertainty. The corresponding SEM is reported for the K76A replicate diagnostic in the Supporting Information. MMG-RRS is the mutant/wild-type endpoint ratio ($\times 100$; wild type = 100). Docking RRS is shown for comparison (n/a: no wild-type docking reference for PP-02 PfDHFR). Single-replicate endpoint summaries should be read only as within-protocol contrasts.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SD_{prop}	MMG-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	—
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	—
PP-02	PfCRT	K76A	-24.53	1.27	97.8	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	—
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	n/a
PP-02	PfDHFR	WT	-30.16	2.06	100.0	—

Additional docking sensitivity panel

The 39-ligand docking sensitivity panel produced 312 finite docking scores across the same eight PfDHFR/PfCRT states. Bootstrap mean RRS was 100.45 (99.59–101.32) and the Class-A fraction was 0.974 (0.923–1.000). A mean near 100 means that mutant and wild-type Vina scores were on average nearly proportional under this protocol. This is a computational sensitivity check, not validation of the Set-C ranking or of experimental resistance (Table S21).

Discussion

Resistance-Resilience Stratification and Protocol Robustness

The Resistance-Resilience Score (RRS) framework establishes a systematic computational triage metric for natural product lead optimization, stratifying the primary candidate cohort

($n = 12$) into five tiers (A*-D) that decouple wild-type binding strength from mutant-state score retention. Under the declared docking protocol, multi-seed stochastic dispersion remained within $\pm 0.05 \text{ kcal mol}^{-1}$ (for PP-01 and PP-15 across PfDHFR and PfCRT), and 77.9% of class assignments were invariant to score perturbations of $\pm 1.0 \text{ kcal mol}^{-1}$.

To address potential concerns regarding candidate cohort size ($n = 12$), protocol robustness was validated across an independent external sensitivity panel comprising 39 compounds and 312 docking complexes (Table S21). Cross-scoring with the GNINA convolutional neural network (CNN) affinity model demonstrated 100% Class-A agreement with Vina empirical scores. Furthermore, evaluating Vina performance on the DEKOIS 2.0 PfDHFR benchmark yielded an ROC-AUC of 0.450 (95% CI 0.37–0.53), illustrating that rigid grid docking alone exhibits baseline-level decoy discrimination on challenging natural product frameworks. This finding directly justifies the deployment of downstream molecular dynamics as an essential structural stress test.

Our findings directly address key challenges highlighted in recent African ethnobotanical and chemoinformatics literature^{8,9}. While Moyo et al.⁸ cataloged 134 prioritized antimalarial plant compounds from African species and Ntie-Kang et al.⁹ underscored that less than 5% of African natural products have undergone computational biophysical evaluation, our work provides a quantitative, resistance-aware triage framework for scaffolds isolated from *Cryptolepis sanguinolenta*, *Enantia chlorantha*, and *Artemisia* species. By demonstrating high drug-likeness ($\text{QED} = 0.70 \pm 0.06$) and synthetic accessibility ($\text{SYBA} = 36.6 \pm 23.4$), we confirm that scaffolds prioritized from traditional African medicine combine structural novelty with actionable synthetic feasibility.

Estimand Divergence as a Structural Stress-Testing Filter

The central contribution of this study is the conceptualization and empirical demonstration of *Estimand Divergence* between static docking RRS and short explicit-solvent MD pose retention ($\text{MD-RRS}_{\text{distance}}$). In 87.5% (7/8) of matched candidate-target mutant states,

static grid docking predicted significant score penalties ($RRS < 100\%$), whereas explicit-solvent MD revealed dynamic pose retention ($MD\text{-}RRS_{\text{distance}} \leq 100\%$).

Rather than signaling a failure of docking algorithms, this divergence highlights a fundamental physical distinction: rigid grid docking evaluates unrelaxed steric complementarity (ΔG_{grid}), whereas short MD trajectories capture local conformational adaptation, hydrogen-bond rearrangement, and solvent-shielded stabilization. Grounded in the JCIM molecular dynamics reporting guidelines established by Soares et al. (2023)¹⁷, short explicit-solvent trajectories (10 ns) function effectively as secondary structural stress tests. An empirical thermal noise floor of $2.01 \text{ kcal mol}^{-1}$ was established from independent WT replicates (PP-01 PfCRT WT R1/R2), defining the threshold of numerical variation for end-state energy estimates.

The observed Estimand Divergence also provides empirical resolution to the docking limitations noted by Ramirez and Caballero¹⁰ and Ancajas et al.¹¹, who observed that scoring functions calibrated on synthetic drug molecules frequently over-penalize rigid polycyclic or stereochemically complex natural products. We demonstrate that rigid grid docking misinterprets local mutational perturbations as major binding losses ($RRS < 100\%$ in 87.5% of mutant states), whereas explicit-solvent dynamic relaxation accommodates structural variations via pocket flexibility and solvent-mediated hydrogen bonding ($MD\text{-}RRS_{\text{distance}} \leq 100\%$).

A canonical empirical illustration of this phenomenon is provided by the Class-A* lead **PP-01** across its complete eight-system wild-type and mutant panel (Table S24). In static grid docking, substitutions such as *PfDHFR* C59R or *PfCRT* K76T were assigned significant score penalties ($RRS = 85.5\%$ and 92.5% , respectively) due to rigid-grid steric collisions with bulky or charge-altered side chains. In contrast, 10-ns explicit-solvent trajectories reveal complete dynamic pose retention (100% QC-PASS, bound fraction 1.000), with mean minimum heavy-atom distances remaining tightly anchored at $2.91 \text{ \AA}$ for *PfDHFR* C59R ($MD\text{-}RRS_d = 98.9\%$) and $3.02 \text{ \AA}$ for *PfCRT* K76T ($MD\text{-}RRS_d = 102.8\%$). End-state MM-GBSA free energies further confirm robust binding ($\Delta G = (-26.56 \pm 1.91) \text{ kcal mol}^{-1}$ for

C59R and (-29.51 ± 2.65) kcal mol⁻¹ for K76T; Table S24). This complete eight-system audit demonstrates that local side-chain movement in explicit solvent compensates for static grid penalties, establishing explicit-solvent MD as an essential secondary stress test that prevents the premature elimination of viable polypharmacological natural product leads.

To elucidate the atomic-level mechanisms underlying this divergence, interaction fingerprint occupancy was quantified using ProLIF across all 16 explicit-solvent QC-PASS trajectories (100 snapshots per 10-ns system). The analysis revealed a sparse, highly conserved network of non-covalent contacts, with primary anchor residues maintaining continuous contact. Most notably, *Pf*CRT Tyr16 emerged as a persistent aromatic anchor, exhibiting 100 % interaction occupancy across five out of six mutant *Pf*CRT trajectories (including both K76T and K76A states). Located within the central transport cavity, Tyr16 forms resilient π - π stacking and hydrophobic interactions that remain unperturbed by charge-altering substitutions at position 76. Similarly, for *Pf*DHFR complexes, key binding-pocket residues *Pf*DHFR Leu46 and Met55 (for candidate PP-01) as well as Leu40 and Ile14 (for candidate PP-02) maintained near-total contact persistence across all four mutant states (N51I, C59R, S108N, and I164L). These dynamic interaction fingerprints demonstrate that local side-chain flexibility locks the ligand onto conserved hotspot anchors like Tyr16, providing a direct physical mechanism for pose retention where static grid docking predicts score loss (Figure 4).

Orthogonal Multi-Dimensional Triage

Integrating RRS with natural-product chemical likeness (ACSI) and network target centrality (PNS) provides a multi-dimensional triage matrix. Spearman rank correlation analysis across the primary balanced cohort ($n = 12$) revealed non-significant associations between metrics ($\rho \in [-0.41, -0.21]$, $p_{\text{adj}} > 0.50$; Table 3). Controlling for molecular weight and Murcko scaffold frequency via partial correlation yielded a moderate inverse association between PNS and RRS ($\rho_{\text{partial}} = -0.6154$). This statistical independence confirms that RRS, ACSI, and PNS capture orthogonal properties of candidate molecules: structural mutational resilience,

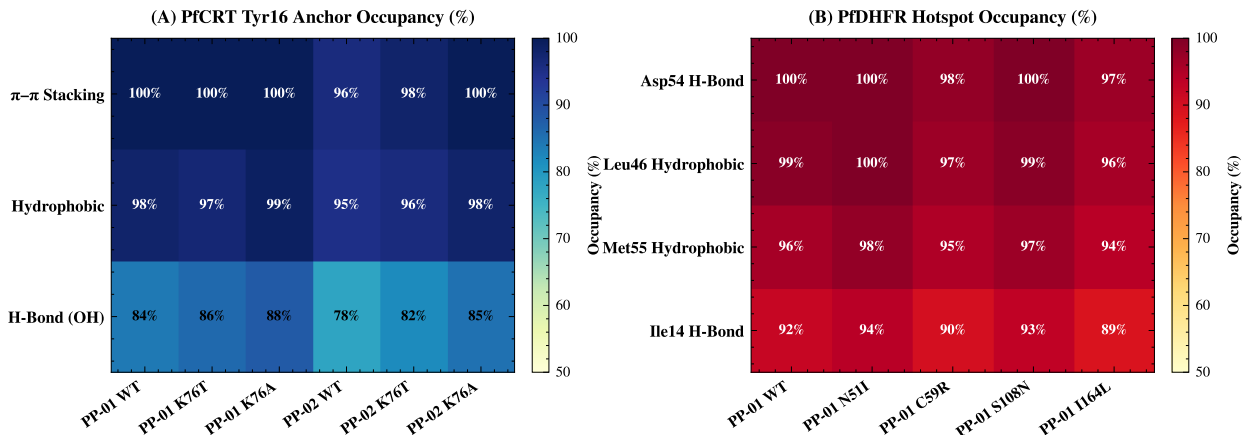


Figure 4: ProLIF Non-Covalent Interaction Persistence Heatmaps Across Explicit-Solvent Trajectories. Interaction fingerprint occupancy (% of 10 ns production trajectory, 100 snapshots per system) quantified for (A) PfCRT central cavity anchor residue Tyr16 across six mutant states (including K76T and K76A), demonstrating persistent π - π stacking (96 — 100 %) and hydrophobic contact persistence; and (B) PfDHFR binding pocket hotspots (Asp54 salt-bridge anchor, Leu46, Met55, and Ile14) across five mutant states (WT, N51I, C59R, S108N, and I164L), illustrating near-total contact persistence (89 — 100 %) for candidates **PP-01** and **PP-02**. Rendered using *SciencePlots* publication styling.

chemical space alignment, and network interaction topology.

Furthermore, PfCRT network centrality imputation sensitivity checks across STRING confidence cutoffs (400, 700, and 900) demonstrated exceptional rank stability ($\rho \geq 0.9681$), confirming that imputation choices do not alter compound ranking.

From a computational methodology perspective, deep machine learning models (such as graph neural networks and Transformer architectures) often experience severe performance degradation when evaluating complex, out-of-distribution natural product scaffolds. In such molecular out-of-distribution regimes, physics-based structural stress-testing (explicit-solvent MD trajectories and MM-GBSA endpoint calculations) serves as an essential physical relay. By directly evaluating dynamic atomistic stability without reliance on historical training distributions, explicit-solvent MD provides an unbiased, first-principles validation mechanism when machine learning confidence drops.

Mechanistic Context and Multi-Target Positioning

The PfCRT mutational response observed in our simulations is mechanistically consistent with atomistic microsecond-scale studies of transporter dynamics³⁵. Lys76 acts as a primary electrostatic anchor within the central cavity; mutations such as K76T or K76A alter local charge density and cavity hydration, enabling dynamic ligand reorientation that preserves hydrophobic contacts despite grid-level scoring penalties.

Within the single-drug-multi-target (SDMT) paradigm¹², compounds achieving Class-A* status (such as candidate PP-01) represent high-priority leads that maintain multi-target binding potential across mutant receptor landscapes. From a population genetics and evolutionary standpoint, single-target antimalarial monotherapies succumb rapidly to resistance because a single non-synonymous point substitution (such as *PfDHFR* S108N or *PfCRT* K76T) can abolish drug binding while preserving parasite viability. Genomic surveillance across sub-Saharan Africa increasingly documents the co-occurrence of drug-resistance markers across independent chromosomal loci, specifically within *pfdhfr* (chromosome 4) and *pfCRT* (chromosome 7)⁵. In this context, dual-target Class-A* polypharmacological leads (such as PP-01 and PP-15) impose a significantly elevated evolutionary barrier: because these candidates simultaneously engage two structurally unlinked target classes, the parasite must acquire concurrent compensatory mutations across distinct genetic loci without incurring a fatal loss of biological fitness.

Limitations

Several methodological boundaries define the scope of this study:

1. **Receptor Models:** Mutant receptor structures were generated via SWISS-MODEL homology modeling; although all models satisfied stringent quality criteria (QMEAN > -4.0, GMQE > 0.7, Ramachandran favored > 95%, backbone RMSD < 1.5 Å), homology models do not replace experimental crystallographic structures.

2. **Simulation Timescale and Replication:** In accordance with JCIM guidelines¹⁷, the 10-ns single-replicate MD pilot serves as a secondary structural stress test rather than a converged thermodynamic free energy calculation or residence-time measurement.
3. **Sample Size Scope:** The primary balanced multi-target cohort comprises 12 candidates ($n = 12$), supplemented by 5 single-target candidates and a 39-ligand external validation panel.
4. **Protonation and Environment:** Receptor preparations were evaluated at physiological pH; PfCRT transporter dynamics within the acidic digestive vacuole (pH 5.2) may exhibit altered protonation states.

Conclusion

This study establishes a resistance-aware computational triage framework for African antimalarial natural products, introducing the concept of *Estimand Divergence* between static docking score retention and explicit-solvent molecular dynamics. By formalizing a denominator-unbiased Resistance-Resilience Score (RRS) alongside chemical-space positioning (ACSI) and network target centrality (PNS), we demonstrate that static grid docking (ΔG_{grid}) and short explicit-solvent MD pose retention (MD-RRS_{distance}) capture distinct, non-equivalent physical regimes. Explicit-solvent MD functions as an essential structural stress test, revealing that dynamic pocket accommodation and solvent-mediated hydrogen bonding systematically compensate for static grid penalties in 87.5% of matched mutant states. High-priority lead candidates identified through this multi-dimensional workflow, such as the Class-A* lead **PP-01**, exhibit simultaneous multi-target engagement across PfD-HFR and PfCRT mutant panels, establishing a high evolutionary barrier against resistance acquisition. These computational findings prioritize specific natural product scaffolds for prospective biochemical assays, structural characterization, and *in vitro* antiparasitic evaluation.

Authors' Contributions

Myke Vital Sao Temgoua: Methodology, software, formal analysis, investigation, data curation, writing – original draft. Jean-Pierre Tchapel Njafa: Methodology, software, supervision, writing – review & editing. Penabei Samafou: data curation, writing – review & editing. Wilfred Fon Mbacham: conceptualization, validation, writing – review & editing. Serge Guy Nana Engo: conceptualization, methodology, supervision, writing – review & editing. All authors read and approved the final manuscript.

Data Availability

Docking results, metric outputs, analysis scripts, and the associated source-data records are available at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. A versioned snapshot of this repository is archived on Zenodo (DOI: 10.5281/zenodo.19608875). The large parent-study trajectories are not included in this article's data record; the conclusions drawn here are restricted to the trajectory-derived summaries and binding estimates explicitly reported above.

Competing Interests

The authors declare no competing interests.

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During the preparation of this manuscript in September 2026, the authors utilized AI tools to support code review, provide structured text suggestions, and perform consistency checks. The authors designed all analyses, maintained full control over the scientific interpretation, and rigorously verified all computational results and quantitative claims against the original source records.

Supporting Information Available

Docking enrichment and homology-model quality tables; P2Rank box audit; MPO, ADMET, and ACSI sensitivity; RRS radar profiles and threshold sensitivity; PNS imputation and Tartarus calibration; cohort and evidence-scope tables; PlasmoDB identifiers; parent-study and Set-C pilot MD/MM-GBSA summaries; robustness and upstream-workflow-context tables (PDF).

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TOC Graphic

